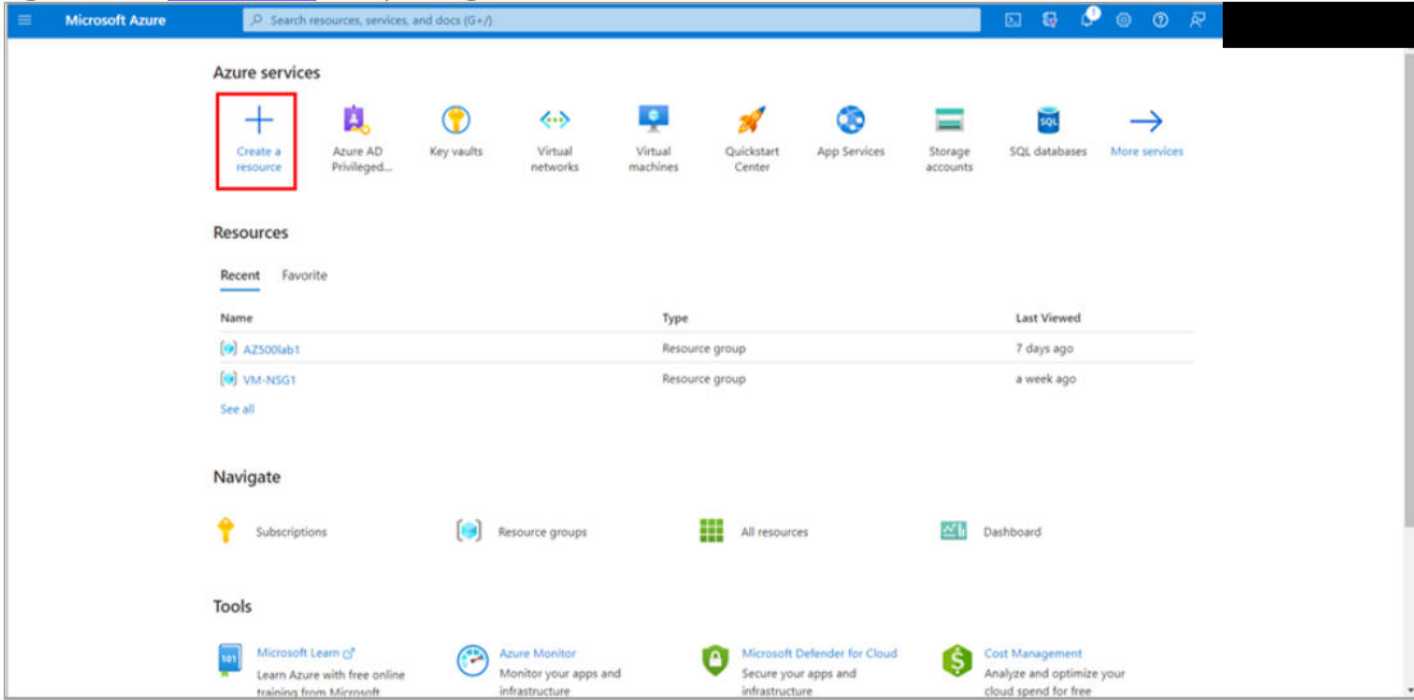


Key Vault project solution

Instructions: List the steps you will perform to complete each task in the system. Ensure to provide screenshots for each step. As a sample, we have filled in the information for the first two steps in *Task 1, Activity 1*. We have used red highlights to show how relevant the screenshot is for the step. However, you do not need to use any highlights in your work. You can add more rows as required.

Task 1: Store a secret in a Key Vault and use it during virtual machine creation.

In this task, you store a secret, which is a password for your Windows virtual machines (VMs).

Activity #	Steps									
Activity 1: Create a Key Vault.	1. Sign in to the Azure portal with your login credentials. Select Create a resource.  <table><tr><th>Name</th><th>Type</th><th>Last Viewed</th></tr><tr><td>AZ500lab1</td><td>Resource group</td><td>7 days ago</td></tr><tr><td>VM-NSG1</td><td>Resource group</td><td>a week ago</td></tr></table> 2. In the Search box, enter Key Vault. From the results list, select Key Vault.	Name	Type	Last Viewed	AZ500lab1	Resource group	7 days ago	VM-NSG1	Resource group	a week ago
Name	Type	Last Viewed								
AZ500lab1	Resource group	7 days ago								
VM-NSG1	Resource group	a week ago								

Microsoft Azure

Search resources, services, and docs (G+J)

Home > Create a resource >

Marketplace

Get Started

Service Providers

Management

Private Marketplace

Private Offer Management

My Marketplace

Favorites

Recently created

Private products

Categories

Security (30)

DevOps (18)

Identity (16)

Analytics (10)

Compute (10)

Key Vault

Pricing : All Operating System : All Publisher Type : All Product Type : All Publisher name : All

Showing 1 to 20 of 80 results for 'Key Vault'. [Clear search](#)

Tile view

**Key Vault**

Microsoft

Azure Service

Safeguard cryptographic keys and other secrets used by cloud apps and services.

Create

**KoçSistem Azure Key Vault Management**

KoçSistem Bilgi ve İletişim Hızın...

Managed Services

Increase security and control over keys and passwords with the support of Azure Key Vault Services!

Create

**Azure Key Vault solution for Sentinel**

Microsoft Sentinel, Microsoft Co...

Azure Application

Azure Key Vault solution for Sentinel

Price varies

Create

**Azure Key Vault Managed HSM**

Microsoft

Azure Service

Safeguard cryptographic keys used by cloud apps and services.

Create

**EJBCA SaaS - With Key Vault Backed CA Keys**

Keyfactor, Inc.

SaaS

EJBCA SaaS. PKI delivered as a service with Azure Key Vault key storage

Starts at **₺388,617.275/month**

Subscribe

**HashiCorp Vault Enterprise**

HashiCorp

SaaS

Create

**Hashicorp Vault on CentOS**

AskforCloud LLC

Virtual Machine

Create

**EJBCA SaaS - With Azure Key Vault Managed HSM**

Keyfactor, Inc.

SaaS

Create

**Vault on Debian 10**

AskforCloud LLC

Virtual Machine

Create

**Vault on Debian 11**

AskforCloud LLC

Virtual Machine

Create

Is Marketplace helpful?

3. Keyvault has been created.

The screenshot displays the Microsoft Azure portal interface. The top navigation bar includes the 'Microsoft Azure' logo, an 'Upgrade' button, a search bar, and a 'Copilot' button. The left sidebar contains a 'Create a resource' button and a list of services including 'Home', 'Dashboard', 'All services', 'FAVORITES', 'Application Templates', 'All resources', 'Resource groups', 'App Services', 'Function App', 'Azure SQL Database', 'Azure Cosmos DB', 'Virtual machines', 'Load balancers', 'Storage accounts', 'Virtual networks', 'Microsoft Entra ID', 'Monitor', 'Advisor', 'Microsoft Defender for Cloud', and 'Cost Management + Billing'.

The main content area shows the 'secret-kv-new' Key Vault overview page. The breadcrumb navigation is 'Home > Ungraded-keyvault'. The page title is 'secret-kv-new' with the subtitle 'Key vault'. There are two tabs: 'Overview' (selected) and 'JSON View'. The 'Overview' tab displays the following information:

- Resource group:** [Ungraded-keyvault](#)
- Location:** East US
- Subscription:** [Azure subscription 1](#)
- Subscription ID:** ae057139-3ee2-4662-a3fb-11c8432a3437
- Vault URI:** <https://secret-kv-new.vault.azure.net/>
- Skus (Pricing tier):** Standard
- Directory ID:** 54047325-4de1-43d6-b137-a1a36fdb06d4
- Directory Name:** Default Directory
- Soft-delete:** [Enabled](#)
- Purge protection:** [Disabled](#)

Below the 'Essentials' section, there are tabs for 'Get started', 'Properties', 'Monitoring', 'Tools + SDKs', and 'Tutorials'. The 'Get started' tab is selected, showing a section titled 'Manage keys and secrets used by apps and services'. The text below this title reads: 'Our recommendation is to use a vault per application per environment (Development, Pre-Production and Production). This helps you to not share secrets across environments and also reduces the threat in case of a breach.'

At the bottom of the page, there are three cards with checkmarks indicating completion status:

- Control access to key vault** (checked)
- Enable logging and set up alerts** (checked)
- Turn on recovery options** (checked)

Activity 2:
Save a secret in Azure Key Vault.

1. Saving a secret in the key vault

Microsoft Azure

Upgrade

Search resources, services, and docs (G+/I)

Copilot

Create a resource

Home

Dashboard

All services

FAVORITES

Application Templates

All resources

Resource groups

App Services

Function App

Azure SQL Database

Azure Cosmos DB

Virtual machines

Load balancers

Storage accounts

Virtual networks

Microsoft Entra ID

Monitor

Advisor

Microsoft Defender for Cloud

Cost Management + Billing

Home > Ungraded-keyvault > secret-kv-new | Secrets

Create a secret

Upload optionsManual

Name * Password

Secret value *

Content type (optional)

Set activation date

Set expiration date

EnabledYes No

Tags0 tags

CreateCancel

2. Made myself the Key Vault Administrator to manage secrets

The screenshot shows the Microsoft Azure portal interface. The top navigation bar includes the Microsoft Azure logo, an Upgrade button, a search bar, and a Copilot button. The left sidebar contains a list of services and favorites. The main content area displays the 'Add role assignment' page for the 'Key Vault Administrator' role. The page has tabs for Role, Members, Conditions, and Review + assign. The Role tab is selected, showing the role name, scope, and a table of members. The scope is '/subscriptions/ae057139-3ee2-4662-a3fb-11c8432a3437/resourceGroups/Ungraded-keyvault/providers/Microsoft.KeyVault/vaults/secret-kv-new'. The members table lists Charlie Chaplin(Guest) with Object ID 2f043ba1-7fc9-4aa4-a2f6-362f02b6a886 and Type User. The Description is 'No description'. At the bottom, there are buttons for 'Review + assign', 'Previous', and 'Next'. A Feedback link is also present.

Home > secret-kv-new | Access control (IAM)

Add role assignment

Role Members Conditions Review + assign

Role Key Vault Administrator

Scope /subscriptions/ae057139-3ee2-4662-a3fb-11c8432a3437/resourceGroups/Ungraded-keyvault/providers/Microsoft.KeyVault/vaults/secret-kv-new

Members	Name	Object ID	Type
	Charlie Chaplin(Guest)	2f043ba1-7fc9-4aa4-a2f6-362f02b6a886	User

Description No description

[Review + assign](#) [Previous](#) [Next](#)

[Feedback](#)

Microsoft Azure

Upgrade

Search resources, services, and docs (G+J)

Copilot

Create a resource

Home

Dashboard

All services

FAVORITES

Application Templates

All resources

Resource groups

App Services

Function App

Azure SQL Database

Azure Cosmos DB

Virtual machines

Load balancers

Storage accounts

Virtual networks

Microsoft Entra ID

Monitor

Advisor

Microsoft Defender for Cloud

Cost Management + Billing

secret-kv-new

secret-kv-new | Secrets

Key vault

Search

Generate/Import

Refresh

Restore Backup

Manage deleted secrets

View sample code

The secret 'Password' has been successfully created.

Name	Type	Status	Expiration date
Password		✓ Enabled	

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Access policies

Resource visualizer

Events

Objects

Keys

Secrets

Certificates

Settings

Monitoring

Automation

Help

Add or remove favorites by pressing Ctrl+Shift+F

Give feedback

Activity 3:
Create a DevTest Lab.

Microsoft Azure

Upgrade

Search resources, services, and docs (G+)

Copilot

Create a resource

Home

Dashboard

All services

FAVORITES

Application Templates

All resources

Resource groups

App Services

Function App

Azure SQL Database

Azure Cosmos DB

Virtual machines

Load balancers

Storage accounts

Virtual networks

Microsoft Entra ID

Monitor

Advisor

Microsoft Defender for Cloud

Cost Management + Billing

Home > DevTest Labs

Dev-ungradedlab

DevTest Lab

Search

«

+ Add

Refresh

Claim any

Delete

Feedback

Forum

Overview

Getting started

Internal support

Resource visualizer

My Lab

- Claimable virtual machines
 - All resources
 - My virtual machines
 - My environments
 - My data disks
 - My secrets

Settings

Upcoming Retirements on Sept 30, 2025:

- Basic Load Balancers and Public IPs: Upgrade existing Basic Load Balancers and Public IPs to Standard SKU. To avoid disruptions post-upgrade, do not rename the load balancer during the upgrade. For more information, see the [official announcement](#) or read our [upgrade by name guidance](#).
- Default Outbound Access for Private IP VMs: Find out how you can [configure your outbound access here](#) for your VMs to ensure continued access.

Essentials

JSON View

Resource group [\(move\)](#)
[Ungraded-keyvault](#)

Status
Ready

Location
East US

Subscription [\(move\)](#)
[Azure subscription 1](#)

Subscription ID
ae057139-3ee2-4662-a3fb-11c8432a3437

My virtual machines

Virtual machines you have added and claimed appear in this list. Choose "claim any" to immediately claim a random virtual machine, or see [Claimable virtual machines](#) to choose among all existing machines available in this DevTest Lab.

Name ↑↓	Status ↑↓	Auto-start ↑↓	Auto-shutdown ↑↓	Base ↑↓
Nothing to display				

Add or remove favorites by pressing Ctrl+Shift+F

Activity 4:
Create a virtual machine
in DevTest Lab.

1. Setting up a VM in DevTest Lab

Basic Settings

Advanced Settings

Tags

User Settings

Virtual machine name *

TheGreatVM

✓

User name *

HxVM

✓

Use a saved secret ⓘ

☐

Password *

.....

✓

Save as default password

☐

Disk and Size

Image Base

Windows 11 Enterprise multi-session + Microsoft 365 Apps, version 22H2 - x64 Gen2

Virtual machine size * ⓘ

Size

Standard_D4s_v5

[Change Size](#)

Allow hibernation ⓘ

☐

OS disk type ⓘ

Standard SSD

▼

Security features

Security type ⓘ

Standard

Trusted launch

Artifacts

Artifacts

Artifact

0 artifact(s) configured

[Add or Remove Artifacts](#)

2. Added chrome as an artifact in the VM as stated in the *Project Requirements*.

⚠️ Applying artifacts on small sized (1 core) Windows VMs may take a longer time or fail in certain cases

Select the artifacts to be installed on the VM. Artifacts will be installed in the order chosen, from top to bottom. [Learn more.](#)

1 artifact(s) configured

Search to filter available artifacts

Available artifacts

Name

Loading...

Selected artifacts

Name

Chrome

Atom

AWS Command Line Interface

Azure Pipelines Agent

Azure Pipelines Agent(s)

Azure Pipelines Deployment Group Agent

Azure PowerShell

Beyond Compare

Chef Client

Chrome

Clone a git repository

Configuration Manager Current Branch

Configuration Manager Technical Preview

Configure WinRM

Create Domain

Activity 5:
Complete the virtual machine deployment.

1. VM deployment **failed** due to an error on Azure’s side.

Microsoft Azure

Upgrade

Search resources, services, and docs (G+/)

Copilot

Create a resource

Home

Dashboard

All services

FAVORITES

Application Templates

All resources

Resource groups

App Services

Function App

Azure SQL Database

Azure Cosmos DB

Virtual machines

Load balancers

Storage accounts

Virtual networks

Microsoft Entra ID

Monitor

Advisor

Microsoft Defender for Cloud

Cost Management + Billing

Home > Ungraded-keyvault

vm-lab | Overview

virtual machine

Search

Connect

Start

Restart

Stop

Hibernate

Artifacts

Claim machine

Unclaim

Delete

Refresh

General

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Operations

Create custom image

Create formula (reusable base)

Artifacts

Auto-shutdown

Auto-start

Redeploy

Expiration date

Settings

Disks

Security

Essentials

Status

Failed

Resource group

dev-ungradedlab984571964000

Virtual network/subnet

DtlDev-ungradedlab/DtlDev-ungradedlabSubnet

IP address or FQDN

10.0.0.5

NAT protocol / Port to connect

Private

Size

Standard_D4s_v5

Operating system

Windows

Base

Windows 11 Enterprise, version 25H2 (x64) (Microsoft Dev Box Compati...

Expiration date

No expiration

Security type

Standard

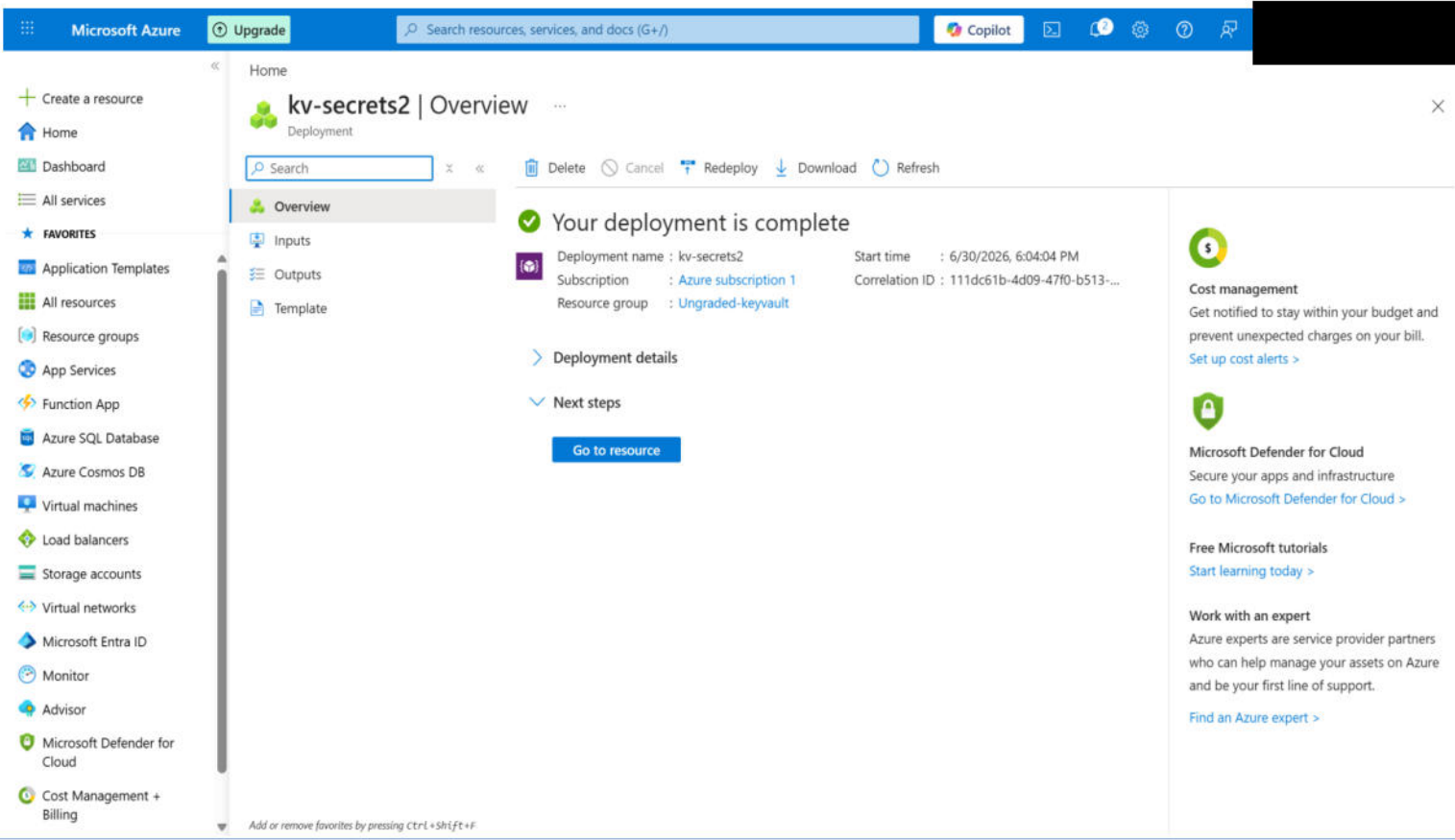
Schedules

Name	Status	Time	Time zone	Day of week
Auto-start	Opted-out			
Auto-shutdown	Opted-out			

Add or remove favorites by pressing Ctrl+Shift+F

Task 2: Create a key in a Key Vault.

In this task, you create a key named *Key1* in a Key Vault named *Key-vaults-secret*.

Activity #	Steps
Activity: Create a key.	1. Creating the key vault named <i>kv-secret2</i> 

2. I made myself the Key Vault Administrator to manage keys

The screenshot shows the Microsoft Azure portal interface. The left sidebar contains navigation options like 'Create a resource', 'Home', 'Dashboard', 'All services', and 'FAVORITES'. The main content area is titled 'kv-secrets2 | Access control (IAM)' and shows the 'Access control (IAM)' tab selected. A warning banner at the top states: 'Action required: 1 user has elevated access in your tenant. You should take immediate action and remove all role assignments with elevated access. In addition, remove any role assignments at root scope that are no longer needed. View role assignments'. Below this, the 'Role assignments' tab is active, showing a table of role assignments for the subscription. The table has columns for Name, Type, Role, Scope, and Condition. It lists two 'Owner' roles and one 'Key Vault Administrator' role, all assigned to the user 'Charlie Chaplin'.

Home > kv-secrets2

kv-secrets2 | Access control (IAM) ☆ ...

Key vault

Search

+ Add Download role assignments Edit columns Refresh Delete Feedback

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Access policies

Resource visualizer

Events

Objects

Keys

Secrets

Certificates

Settings

Monitoring

Automation

Help

Check access Role assignments Roles Deny assignments

Number of role assignments for this subscription ⓘ

7 4000

Search by name or email

Type: All Role: All Scope: All scopes Group by: Role

All (4) Job function roles (1) Privileged administrator roles (3)

Name ↑↓	Type ↑↓	Role ↑↓	Scope ↑↓	Condition ↑↓
Owner (2)				
☐ Charlie Chaplin 2f043ba1-7fc9-4aa...	User	Owner	Subscription (Inherite...	None
☐ Charlie Chaplin 2f043ba1-7fc9-4aa...	User	Owner	Subscription (Inherite...	None
Key Vault Administrator (1)				
☐ Charlie Chaplin 2f043ba1-7fc9-4aa...	User	Key Vault Administrator	This resource	None
User Access Administrator (1)				
☐ Charlie Chaplin				

Add or remove favorites by pressing Ctrl+Shift+F

3. Creating the key with rotation policy for auto-rotation of the key's value.

The screenshot displays the Microsoft Azure portal interface for creating a new key in a Key Vault. The main pane shows the 'Create a key' wizard with the following options:

- Options:** Generate
- Name:** KeyToHeaven
- Key type:** RSA (selected), EC
- RSA key size:** 2048 (selected), 3072, 4096
- Set activation date:** ☐
- Set expiration date:** ☐
- Enabled:** Yes (selected), No
- Tags:** 0 tags
- Set key rotation policy:** Not configured
- Confidential Key Options:**
 - Exportable:** ☐
 - Immutable:** ☐

The 'Rotation policy' sidebar is open, showing the following settings:

- Expiry time:** 30 days
- Rotation:**
 - Enable auto rotation:** Enabled (selected), Disabled
 - Rotation option:** Automatically renew at a given time after c... (selected)
 - Rotation time:** 10 days
- Notification:**
 - Notification option:** Notify at a given time before expiry
 - Notification time:** 30 days

The 'Create' button is visible at the bottom of the main pane, and the 'OK' button is visible at the bottom of the 'Rotation policy' sidebar.

4. The key is created successfully.

The screenshot shows the Microsoft Azure portal interface. The left sidebar contains navigation options like 'Create a resource', 'Home', 'Dashboard', 'All services', and 'FAVORITES'. The main content area is titled 'kv-secrets2 | Keys' and shows a table with the following data:

Name	Status	Expiration date
KeyToHeaven	✓ Enabled	

Two notifications are displayed on the right side of the screen:

- Updating the key 'KeyToHeaven'. The key 'KeyToHeaven' has been successfully updated.
- Creating the key 'KeyToHeaven'. The key 'KeyToHeaven' has been successfully created.